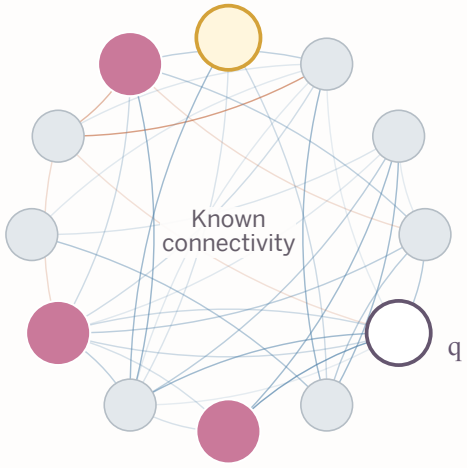
Predictive risk and recording decisions

in partially observed neural networks

Nominal versus prior-averaged local risk for stimulus-response prediction



12 simulated units · M=3

Uncertain gains and biases

Blue: positive; orange: negative connections

- Recorded node
- Stimulated node
- Response readout (one shown)

1 Recording agreement

40 independent networks

Single-response designs

68% 82 of 120 designs identical recording sets

Shared three-response designs

65% 26 of 40 designs identical recording sets

same recording set

2 Response error

Common MAP fit and budget
Same 40 networks

Single response

−3.7% [−18.5, +12.1]

Shared three-response

+2.7% [−3.0, +11.5]

−20 0 15

Mean-MSE change vs nominal (%)

97.5% paired network-cluster CI

3 Risk forecasts

Fixed data and point prediction
40 new independent networks

19 of 40 favour prior averaging; 21 favour nominal

$S(v, e) = \log v + e^2/v$

v: MSE forecast; e: response error

One network accounts for 64% of the summed score difference

Forecast-score difference

−0.46 [−1.16, +0.02]

−1.2 −0.6 0.0

Prior-averaged minus nominal

95% paired network-cluster CI

Integrated stimulus response

$$q = \int \Delta r(t)\, dt$$

$$\Delta r(t) = r_{\text{stim}}(t) - r_{\text{control}}(t)$$

Identical recording sets and data yield identical predictions under the common MAP fit.
All primary comparison intervals include zero.

Exploratory 100-unit rank-two family (8 independent networks): 4/8 setting-specific 95% intervals exclude zero, all favouring prior averaging. Intervals are unadjusted; one point estimate favours nominal.